

不定积分的基本概念

一、研究的问题： $(\ ?)'_x = f(x)$.

二、计算方法：

1、简单积分：不改变积分变量，找出原函数

想法：合并，拆项，降幂，贴近公式表现形式

2、第一换元法：改变积分变量表现形式，不改变自变量

$$\int f(\varphi(t))\varphi'(t)dt = \int f(\varphi(t))d\varphi(t) = F(\varphi(t)) + C$$

想法

简单部分先积分，“凑”复杂部分的自变量

何为“复杂”？

3、第二换元法：同时改变积分变量表现形式、自变量

$$\int \boxed{f(x)} dx \underset{x=\varphi(t)}{=} \int f(\varphi(t)) d\varphi(t) = \int \boxed{f(\varphi(t))\varphi'(t)} dt$$

形式复杂，难于计算



合并简化，形式简单，
特征明显，易于计算

本质：第二换元法，通过变量代换、求导合并简化被积函数
具体的应用一般是“消除”根式。

题型 $\int f(x)dx$: $f(x)$ 含有 $\sqrt[n]{ax+b}$

题型 $\int f(x)dx$: $f(x)$ 含有 $\sqrt{ax^2+bx+c}$

$$\int f(x)g(x)dx = \left\{ \begin{array}{l} \text{没有公式} \\ \text{有想法:} \end{array} \right. \left\{ \begin{array}{l} \text{简单积分} \\ \text{换元积分} \end{array} \right.$$

$$\int x \sin x dx = ? \quad \text{无能为力! 寻找新的方法!}$$

四、分部积分:

$$\begin{aligned} \text{推导过程:} \quad (uv)' &= uv' + u'v & uv' &= (uv)' - u'v \\ & & \int uv' dx &= uv - \int u'v dx \end{aligned}$$

$$\text{公式内容:} \quad \int uv' dx = uv - \int u'v dx$$

本质目的: 转化! 难算的积分转化为简单的积分

四、分部积分：

公式内容： $\int uv' dx = uv - \int u' v dx$

本质目的： 转化！ 难算的积分转化为简单的积分

例 $\int x \sin x dx$

$$u = x, \quad v' = \sin x \Rightarrow u' = 1, \quad v = -\cos x$$

$$\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x + c$$

公式内容: $\int uv' dx = uv - \int u' v dx$

计算过程:

$$\int uv' dx = \int u dv = uv - \int v du = uv - \int u' v dx$$

具体的积分: $u = ?$, $v' = ?$

例 $\int x \arctan x dx$

$$= \frac{1}{2} \int \arctan x dx^2 = \frac{1}{2} (x^2 \arctan x - \int x^2 d \arctan x)$$

$$= \frac{1}{2} (x^2 \arctan x - \int x^2 \cdot \frac{1}{1+x^2} dx)$$

$$\int uv'dx = \int u dv = uv - \int v du = uv - \int u'v dx$$

例 $\int x \sin x dx$

$$= \frac{1}{2} \int \sin x dx^2 = \frac{1}{2} (x^2 \sin x - \int x^2 d \sin x)$$

$$= \frac{1}{2} (x^2 \sin x - \int x^2 \cos x dx)$$

解 $\int x \sin x dx = - \int x d \cos x$

$$= -x \cos x + \int \cos x dx = -x \cos x + \sin x + c$$

总结规律;

选择V的
有趣的想法:

能积分的函数v

$$\int x \arctan x dx$$

都能积分, 导函数更简单的是u

$$\int x e^x dx$$

都能积分, 导函数难度程度不变

$$\int e^x \sin x dx$$

二次分部积分, 辟古程

$$\int uv' dx = \int u dv = uv - \int v du = uv - \int u' v dx$$

题型:

常规计算;

递推关系;

抵消;

例1 $\int x^2 e^x dx$

常见同类题型：

$$\int x^2 \sin x dx$$

$$\int x^2 \arcsin x dx$$

$$\int x^2 e^x dx$$

$$\int x^2 \ln x dx$$

例2 $\int \ln x dx$

$$x \ln x - \int x d \ln x$$

例3 $\int \cos(\ln x) dx$ $u dv$

$$= x \cos(\ln x) - \int x d \cos(\ln x)$$

$$= x \cos(\ln x) + \int x \sin(\ln x) \frac{1}{x} dx$$

$$= x \cos(\ln x) + \int \sin(\ln x) dx$$

$$= x \cos(\ln x) + x \sin(\ln x) - \int x d \sin(\ln x)$$

$$- \int x \cos(\ln x) \frac{1}{x} dx$$

$$2 \int \cos(\ln x) dx$$

$$= x \cos(\ln x) + x \sin(\ln x)$$

$$\int \cos(\ln x) dx$$

$$= \frac{x}{2} [\cos(\ln x) + \sin(\ln x)] + c$$

例4 $\int e^{ax} \cos(bx) dx$

$$\text{原式} = \frac{1}{a} \int \cos(bx) d e^{ax}$$

$$= \frac{1}{a} \left[e^{ax} \cos(bx) - \int e^{ax} d \cos(bx) \right]$$

$$+ b \int e^{ax} \sin(bx) dx$$

$$+ \frac{b}{a} \int \sin(bx) d e^{ax} \quad \int e^{ax} d \sin(bx)$$

$$\frac{b}{a} \left[e^{ax} \sin(bx) - b \int e^{ax} \cos(bx) dx \right]$$

例5 $I_n = \int \frac{1}{(1+x^2)^n} dx$

$$I_n = \frac{x}{(1+x^2)^n} - \int x d \frac{1}{(1+x^2)^n}$$

$$= \frac{x}{(1+x^2)^n} + \int x \frac{2nx}{(1+x^2)^{n+1}} dx$$

$$= \frac{x}{(1+x^2)^n} + 2n \int \frac{x^2 + 1 - 1}{(1+x^2)^{n+1}} dx$$

$$= \frac{x}{(1+x^2)^n} + 2n \int \frac{1}{(x^2+1)^n} dx - 2n \int \frac{1}{(1+x^2)^{n+1}} dx$$

$$= \frac{x}{(1+x^2)^n} + 2n I_n - 2n I_{n+1}$$

例6 $I_n = \int \sin^n x dx$

$$I_n = \int \sin^n x dx = x \sin^n x - \int x d\sin^n x = x \sin^n x - \int \underset{\Delta}{n} x \sin^{n-1} \cos x dx$$

递推必同型.

$$\begin{aligned} \text{例} \quad I_n &= \int \sin^n x dx = -\int \sin^{n-1} x d\cos x = -\cos x \sin^{n-1} x + \int \cos x d\sin^{n-1} x \\ &= -\cos x \sin^{n-1} x + \int (n-1) \cos x \sin^{n-2} x dx \quad \cos x = 1 - \sin^2 x \\ &= -\cos x \sin^{n-1} x + \int (n-1) \sin^{n-2} x dx + \int (n-1) \sin^n x dx \end{aligned}$$

$$I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2} + (n-1) I_n$$

$$(n-2) I_n = \cos x \sin^{n-1} x - (n-1) I_{n-1}$$

$$I_n = \frac{1}{n-2} \cos x \sin^{n-1} x - \frac{n-1}{n-2} I_{n-2}$$

$$\text{例7} \quad \int \ln[(x+a)^{x+a} (x+b)^{x+b}] \frac{1}{(x+a)(x+b)} dx$$

$$= \int \frac{(x+a) \ln(x+a) + (x+b) \ln(x+b)}{(x+a)(x+b)} dx$$

$$= \int \frac{\ln(x+a)}{x+b} dx + \int \frac{\ln(x+b)}{x+a} dx$$

$$= \int \frac{\ln(x+a)}{\cancel{x+b}} dx + \int \ln(x+b) d \ln(x+a)$$

$$+ \ln(x+b) \ln(x+a) - \int \frac{\ln(x+a)}{\cancel{x+b}} dx$$

$$= \ln(x+a) \ln(x+b) + c$$

△: 碰到不会搞的莫慌,
先算算, 万一能消掉呢!

例8 $\int e^{\sin x} \left(x \cos x - \frac{\sin x}{\cos^2 x} \right) dx$ 多3个x

$$= \int e^{\sin x} x \cos x dx - \int e^{\sin x} \frac{\sin x}{\cos^2 x} dx$$

$$= \int x de^{\sin x} + \int e^{\sin x} \frac{1}{\cos^2 x} d\cos x$$

$$(e^{\sin x})' = \cos x e^{\sin x}$$

$$= \int x de^{\sin x} - \int e^{\sin x} d\frac{1}{\cos x}$$

$$= xe^{\sin x} - \int e^{\sin x} dx - \frac{e^{\sin x}}{\cos x} + \int \frac{1}{\cos x} de^{\sin x}$$

$$= xe^{\sin x} - \int e^{\sin x} dx - \frac{e^{\sin x}}{\cos x} + \int e^{\sin x} dx$$

$$= xe^{\sin x} - \frac{e^{\sin x}}{\cos x} + C$$

$$\text{例9} \quad \int \left(1 + x - \frac{1}{x}\right) e^{x + \frac{1}{x}} dx$$

$$= \int e^{x + \frac{1}{x}} dx + \int x \left(1 + \frac{1}{x^2}\right) e^{x + \frac{1}{x}} dx$$

$$= \int e^{x + \frac{1}{x}} dx + \int x e^{x + \frac{1}{x}} d\left(x + \frac{1}{x}\right)$$

$$= \quad + \int x de^{x + \frac{1}{x}}$$

$$= \quad + x e^{x + \frac{1}{x}} - \int e^{x + \frac{1}{x}} dx + c$$

$$= x e^{x + \frac{1}{x}} + c$$

例10 $\int \sec^3 x dx$

$$= \int \sec x d \tan x$$

$$= \sec x \tan x - \int \tan x d \sec x$$

$$- \int \tan x \sec x dx$$

$$- \int \sec^3 x - \sec x dx$$

另 $\int \sec^3 x dx = \int \frac{1}{\cos^3 x} dx = \int \frac{\cos x}{\cos^4 x} dx$

$$= \int \frac{1}{(1 - \sin^2 x)^2} d \sin x \quad \text{令 } t = \sin x$$

$$\int \frac{1}{(1 - t^2)^2} dt = \int \frac{1}{(1 - t)^2 (1 + t)^2} dt \quad \text{按有理分式积分算}$$

五、有理分式积分积分：

代数学定理

$$\begin{aligned} & \frac{P_{11}(x)}{(x+1)^2(x-2)^4(x^2+2x+5)^3} \\ &= \frac{A_1}{x+1} + \frac{A_2}{(x+1)^2} \\ &+ \frac{B_1}{x-2} + \frac{B_2}{(x-2)^2} + \frac{B_3}{(x-2)^3} + \frac{B_4}{(x-2)^4} \\ &+ \frac{ax+b}{x^2+2x+5} + \frac{a_1x+b_1}{(x^2+2x+5)^2} + \frac{a_2x+b_2}{(x^2+2x+5)^3} \end{aligned}$$

$$\begin{aligned}
& \int \frac{P_{11}(x)}{(x+1)^2(x-2)^4(x^2+2x+5)^3} dx \\
&= \int \frac{A_1}{x+1} dx + \int \frac{A_2}{(x+1)^2} dx \\
&+ \int \frac{B_1}{x-2} dx + \int \frac{B_2}{(x-2)^2} dx + \int \frac{B_3}{(x-2)^3} dx + \int \frac{B_4}{(x-2)^4} dx \\
&+ \int \frac{ax+b}{x^2+2x+5} dx + \int \frac{a_1x+b_1}{(x^2+2x+5)^2} dx + \int \frac{a_2x+b_2}{(x^2+2x+5)^3} dx
\end{aligned}$$

积分类型：

$$\int \frac{1}{x+a} dx; \quad \int \frac{1}{(x+a)^n} dx$$

第=换元法

↖

$$\int \frac{ax+b}{x^2+2x+5} dx; \quad \int \frac{a_1x+b_1}{(x^2+2x+5)^n} dx$$

应用：含三角函数积分转换为有理分式积分积分：

$$\int R(\sin x, \cos x) dx$$

$$t = \tan \frac{x}{2} \quad x = 2 \arctan t \quad dx = \frac{2t}{1+t^2} dt$$

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{2t}{1+t^2}; \quad \cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{1-t^2}{1+t^2}$$

$$\int R(\sin x, \cos x) dx = \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2t}{1+t^2} dt$$

例 8 求 $\int \frac{1}{1 + \sin x + \cos x} dx$.

解 令 $\tan \frac{x}{2} = t$, 则 $x = 2 \arctan t, dx = \frac{2dt}{1+t^2}$, 于是

$$\sin x = \frac{2t}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}$$

$$\begin{aligned} \int \frac{1}{1 + \sin x + \cos x} dx &= \int \frac{\frac{2}{1+t^2}}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} dt = \int \frac{1}{1+t} dt = \ln |1+t| + C \\ &= \ln \left| 1 + \tan \frac{x}{2} \right| + C \end{aligned}$$

$$\int \left[\frac{f(x)}{f'(x)} - \frac{f^2(x) f''(x)}{[f'(x)]^3} \right] dx$$

解: $\int \left[\frac{f(x)}{f'(x)} - \frac{f^2(x) f''(x)}{[f'(x)]^3} \right] dx$

$$= \int \frac{f(x)}{f'(x)} dx - \int \frac{f^2(x)}{[f'(x)]^3} d[f'(x)]$$

$$= \int \frac{f(x)}{f'(x)} dx + \frac{1}{3} \int f^2(x) d \left[\frac{1}{[f'(x)]^2} \right]$$

$$= \int \frac{f(x)}{f'(x)} dx + \frac{1}{2} \frac{f^2(x)}{[f'(x)]^2} - \frac{1}{2} \int \frac{2f(x) f'(x)}{[f'(x)]^3} dx$$

$$= \frac{1}{2} \frac{f^2(x)}{[f'(x)]^2} + C$$

另法: =

$$\int \left[\frac{f(x)}{f'(x)} - \frac{f^2(x) f''(x)}{[f'(x)]^3} \right] dx$$

$$= \int \frac{f(x)}{f'(x)} \left[1 - \frac{f(x) f''(x)}{[f'(x)]^2} \right] dx$$

$$= \int \frac{f(x)}{f'(x)} \frac{[f'(x)]^2 - f(x) f''(x)}{[f'(x)]^3} dx$$

$$= \int \frac{f(x)}{f'(x)} \frac{[f'(x)]' f'(x) - f(x) [f'(x)]'}{[f'(x)]^3} dx$$

$$= \int \frac{f(x)}{f'(x)} d \frac{f(x)}{f'(x)}$$

$$= \frac{1}{2} \frac{f^2(x)}{[f'(x)]^2} + C$$

例 $\int \frac{1}{x\sqrt{1+x^4}} dx$ 分析: 第=换元法, 分部积分不 okAY! 只有第一换元法

$$\int \frac{1}{x\sqrt{1+x^4}} dx \quad \varphi(x) \text{ ???}$$

例: $\int \frac{1}{x\sqrt{1+x^4}} dx = \int \frac{\varphi'(x)}{\varphi(x)x\sqrt{1+x^4}} dx$

① $\varphi(x) = x^2$

$$\int \frac{2x}{2x^2\sqrt{1+x^4}} dx$$

$$= \int \frac{1}{2x^2\sqrt{1+x^4}} dx^2$$

令 $t = x^2$

$$\text{原} = \int \frac{1}{2t\sqrt{1+t^2}} dt$$

令 $t = \tan m$ 原式 = $\int \frac{\sec^2 m}{2 \tan m \sec m} dm$

$$= \frac{1}{2} \int \csc m \, dm$$

$$= \frac{1}{2} \ln |\csc m - \cot m| + C$$

② $\varphi(x) = x^4$

$$\int \frac{4x^3}{4x^4\sqrt{1+x^4}} dx$$

$$= \int \frac{dx^4}{4x^4\sqrt{1+x^4}}$$

$$= \int \frac{dt}{4t\sqrt{1+t^2}}$$

与左侧同理



谢 谢

欢迎提问及讨论

